Human Computer Interaction

Lecture 01 – Short Note

Human-computer interaction (HCI) is a "discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them"

- **The Goals of HCI:**
    1. **Create usable software.**
    2. **Enhance the usability of existing products.**
    3. **Identify problems and tasks that can be addressed with software products.**

HCI is an interdisciplinary field.

Evolution of HCI interfaces:

1. **Interfaces at the Hardware level**
2. **Interfaces at the Programming level**
3. **Interfaces at the Terminal level**
4. **Interfaces at the Interaction dialogue level**
5. **Interfaces at the work setting**
6. **Interface becomes pervasive.**

User experience

User experience" encompasses all aspects of the end-user's interaction with the company, its services, and its products. User experience focuses on having a deep understanding of users, what they need, what they value, their abilities and also their limitations.

User experience describe 7 factors:

- **Useful**
- **Usable**
- **Findable**
- **Credible**
- **Desirable**
- **Accessible**

- **Valuable**

User-centered design

User-centered design is an iterative design process in which designers focus on the users and their needs in each phase of the design process.

User centered Development Process

1. **Understand context of use.**
2. **Specify user requirements.**
3. **Design solutions.**
4. **Evaluate against requirements.**

Advantages of user centered development

1. **Leads o increased sales and lower costs incurred by customer services.**
2. **Leads to safer products.**
3. **By focusing on all users of a products, designers can recognize the diversity of cultures and human values through UCD.**

Contexts for HCI

- **In HCI studies, the context describes the actual conditions under which the software system is used.**

1. **User context**
2. **Time context**
3. **Physical context**
4. **Computing context**

User Context

- **The user context (also known as *personal context*) represents information about the end-user, which interacts with the system.**
- **This includes information such as:**

user profile (age, preferences)

user's location (absolute position, indoors, outdoors) and orientation

  - **nearby objects**
  - **people nearby**
  - **social situation.**

Time context

- **The time context covers relevant information related to time such as absolute time, date, day of the week and season.**

Physical context

- **The physical context includes everything which is measurable in the environment of the system with which the user interacts.**
- **This includes temperatures, noise levels, lighting situations, traffic conditions, etc.**

Computing Context

- **The computing context contains everything related to computational resources.**
- **This can include things such as available networks, network bandwidth, communication costs and nearby computational resources such as printers or fax machines.**

Social Issues Influencing HCI design and use.

1. **Privacy and Security Issues**
2. **Education and Computer Literacy Level**
3. **Business Needs**
4. **Gender Differences**
5. **Age Difference**
    a) **Children**
    b) **Elderly**
6. **Government Rules, Needs, Policies**
7. **Technology Diversity /Technological innovation**
8. **Emergencies**
9. **Personal/ Individual needs**
10. **Cultural differences**
11. **Human diversity**

Human Diversity

- **Humans could be diverse, based on their abilities, disabilities, age etc.**

Disabilities are of different types:

- **Visual disabilities**
- **Auditory disabilities**
- **Motor disabilities**
- **Cognitive disabilities**

Visual disabilities

Long-sightedness, blindness, colorblindness, are all forms of visual disabilities you need to cater for in your design.

Designing for blindness and low vision

- **Braille keyboards**
- **Special speech software that reads Web pages and other documents aloud.**
- **Screen magnifiers that fit over a display to magnify the entire screen.**
- **Avoid the lines / small symbols**

Auditory disabilities

- **Auditory disabilities affect the hearing and come in varying degrees of severity, up to and including total deafness.**

Designing for Auditory disabilities

- **Documents and screens, you design include access to written versions of the audio material.**
- **Offer transcriptions for audio files. Hearing-impaired users can't use software to read voices. So, help them out and include a transcript.**
- **Offer captions in videos for the hearing impaired.**

Motor disabilities

- **Problems with the mobility and use of the hands and arms thus making the use of hardware of computers impossible.**

Designing for Motor disabilities

- **Use speech inputs (speech recognition) rather than keyboard inputs.**

Sticky Keys: Use of Keyboard from one hand.

Eye Tracking devices.

Adaptive keyboards

Mouth sticks

Head wands

Sip and puff Switch

Oversized Trackball Mouse

Cognitive disabilities

- **limitations in mental functioning and in skills such as communicating, taking care of him or herself, and social skills.**

Designing for Cognitive disabilities

- **Focus on readable content.**
- **The simpler the language, the easier it will be to read for learning-impaired users.**

Browsers specifically designed for people with disabilities:

1. **Braillesurf**
2. **BrookesTalk**
3. **EIAD**
4. **EMACSPEAK**
5. **HomePage Reader**
6. **MultiWeb**

The Impact of HCI

- **Productivity Increases**
- **Reduce training, workshop, etc.**
- **Customer Satisfaction**
- **Good quality and good product.**

Lecture 02- Short Note

Design Thinking

- **It is a design methodology that provides a solution-based approach to solving problems.**
- **In which we seek to**
    - **understand the user**
    - **challenge assumptions**
    - **redefine problems**
    - **attempt to identify alternative strategies**
    - **solutions that might not be instantly apparent with our initial level of understanding.**

The five phases of Design Thinking:

1. **Empathize – with your users**
2. **Define – your users' needs, their problem, and your insights**
3. **Ideate – by challenging assumptions and creating ideas for innovative solutions**
4. **Prototype – to start creating solutions**
5. **Test – solutions**

- **The five phases, stages, or modes are not always sequential.**
- **They do not have to follow any specific order**
- **can often occur in parallel and repeat iteratively.**
- **should not understand the phases as a hierarchal or step-by-step process.**

Non-linear Nature of Design Thinking

- **The five stages are not always sequential**
- **They do not have to follow any specific order and they can often occur in parallel and be repeated iteratively.**

Participatory Design

- **Participatory design is a process that involves developers, business representatives, and users working together to design a solution.**

Goals of Participatory Design

- **Designers and users narrowed**
- **Mutual learning**
- **Users highly involved in design process**
- **Handle conflicting goals between**
- **Workers/users and management**

Why involve users?

- **Improve the knowledge upon which systems are built**
- **Enable workers to develop realistic expectations**
- **Reduce resistance to change**
- **Members participate in decisions that affect their work**

Level of user's involvement

- **Documented studies**
- **Source for data gathering**

questionnaires

observations

interviews

ethnography

- **Part of design team**

Role of Designers

- **Coordinate activities**
- **Facilitate discussion**
- **Prepare materials**
- **Advocate solutions**

Participatory Design Terms

- **User-Centered Design**
- **Human-Centered Systems**
- **Participatory Design**
- **Contextual Design**

Participatory Design – benefits

- **Gives users a voice in the design process.**
- **Enables technical and non-technical participants to contribute equally.**
- **Shifts the focus from purely technical requirements and issues towards the needs of the business and users.**
- **A forum for developers to meet, work with, and understand their users.**
- **A forum for identifying issues and assigning them to people for resolution.**
- **A forum to specify Design Goals that can be used for current and future products or versions.**
- **Enables a team to rapidly design, evaluate and iterate design approaches.**

Limitations of Participatory Design

- **Close collaboration between users and developers**
- **Involvement of users**
- **Not all systems are workplace-based**
- **PD ideology must be adapted for dealing with variations The design**

Behavioral Science

Behavioral science is the study of human behaviors dominantly spanning the disciplines of behavioral economics, cognitive neuroscience, sociology, and anthropology.

Cognition

- **Cognition is "the mental action or process of acquiring knowledge and understanding through thought, experience, and the senses“**
- **Cognition can also be described in terms of cognitive processes**

Examples include:

- **– Attention**
- **– Perception and recognition**
- **– Memory**
- **– Learning**
- **– Reading, speaking and listening**
- **– Problem-solving, planning, reasoning and decision-making**

Lecture 03- Short Note

User research techniques

- **Understand the user - user research**
- **Personas**
- **Storyboarding**
- **Contextual Inquiry**
- **Scenarios**

User research

- **Understand the user's needs, behaviors, experience and motivations.**

How to conduct user research

1. **Goal: why are we doing this research, what are we hoping to get out of it.**
2. **Research questions: what are specific questions we are interested in.**
3. **Method: what specific method we want to use to learn about the questions.**
4. **Participants: what's the user profile we are looking for.**
5. **Protocol: how we want to conduct the session.**

User experience research methods

1. **Attitudinal vs. Behavioral**
2. **Qualitative vs. Quantitative**
3. **Context of Use**

Persona

- **A representation of a customer type. It can reflect your customer type and their preferences.**
- **There are 4 different perspectives of personas:**

Goal-directed –focuses on what the user wants

Role-based –focuses on behavior of the user

Engaging persona –revolves the understanding of fictitious people and their interaction with a product

Fictional persona –comes from the user experience design team based on past interactions with users

Storyboard

- **Storyboards are illustrations that represent shots that ultimately represent a story.**
- **Basically, it's a sequential art, where images are arrayed together to visualize the story.**
- **Stories are an effective and inexpensive way to capture, relate, and explore experiences in the design process.**

Contextual Inquiry

- **By seeing what users do in context, it is easier to see how our products are really used.**
- **Contextual interviews are based on four guiding principles:**
    - **Context**
    - **Partnership**
    - **Mutual interpretation**
    - **Focus**
- **There are three phases to the contextual interview when it is carried out with the user:**
    - **Introduction**
    - **The Main Body**
    - **The Wrap Up**

The Advantages of Contextual interviewing

- **The ability to reveal information and understanding that users might not be aware**
- **The accuracy and detail of information – observing users in their natural environment tends to lead to very accurate information**
- **The flexibility of the method – contextual research can be carried out wherever a user operates**

Scenarios

- **A user scenario is the fictitious story of a user's accomplishing an action or goal via a product.**
- **Scenarios describe the stories and context behind why a specific user or user group comes to your site.**
- **Scenarios are usable in design and the Usability Testing**

What to consider when writing scenarios:

- **Who is the user?** Use the personas that have been developed to reflect the real, major user groups coming to your site.
- **Why does the user come to the (site)?** Note what motivates the user to come to the site and their expectations upon arrival, if any.
- **What goals does he/she have?** Through task analysis, you can better understand the what the user wants on your site and therefore what the site must have for them to leave satisfied.
- **How can the user achieve their goals on the site?** Define how the user can achieve his/ her goal on the site, identifying the various possibilities and any potential barriers

Lecture 04- Short Note

Design for different interactions

- **Ideation**
- **Information architecture**
- **Interface types**
- **Interfaces for interactions**

Ideation

- **The process of generating ideas is referred to as Ideation.**

Golden rules of ideation

- **There are no bad ideas**
- **Capture everything**
- **Go for Hybrid Brainstorming**
- **Quantity over Quality**

Ideation techniques

1. **Brainstorming**
2. **Brainwriting**
3. **Mind Mapping**
4. **Worst Possible idea**
5. **SCAMPER**

Information Architecture

- **Information architecture (IA) is a science of organizing and structuring content of the websites, web and mobile applications, and social media software.**

Principles of Information Architecture

1. **Objects**
2. **Choices**
3. **Disclosure**
4. **Exemplars**
5. **Front doors**
6. **Multiple classification**
7. **Focused navigation**
8. **Growth**

The value for the user

- **Known-item seeking:**
- **Exploratory seeking:**
- **Exhaustive research:**
- **Re-finding:**

The value for the Business

- **Employee Productivity**
- **Sales And Reputation**
- **Acquiring New Members**
- **Reducing Marketing Costs**
- **Reputation And SEO Ranking**
- **Reducing The Cost Of Live Help And Support Documentation**

Important questions for interaction designers

- **What can a user do with their mouse, finger, or stylus to directly interact with the interface? T**
- **What about the appearance (color, shape, size, etc.) gives the user a clue about how it may function?**
- **Do error messages provide a way for the user to correct the problem or explain why the error occurred?**
- **What feedback does a user get once an action is performed?**
- **Are the interface elements a reasonable size to interact with?**
- **Are familiar or standard formats used?**

Interaction Design

- **The goal of interaction design is to create products that enable the user to achieve their objective(s) in the best way possible.**

Interaction types

- **Instructing**
  - **issuing commands and selecting options**
- **Conversing**
  - **interacting with a system as if having a conversation**
- **Manipulating**
  - **interacting with objects in a virtual or physical space by manipulating them**
- **Exploring**
  - **moving through a virtual environment or a physical space**

Advantages of DM

- **Novices can learn the basic functionality quickly**
- **Experienced users can work extremely rapidly to carry out a wide range of tasks, even defining new functions**
- **Intermittent users can retain operational concepts over time**
- **Error messages rarely needed**
- **Users can immediately see if their actions are furthering their goals and if not do something else**
- **Users experience less anxiety**
- **Users gain confidence and mastery and feel in control**

Disadvantages of DM

- **Some people take the metaphor of direct manipulation too literally**
- **Not all tasks can be described by objects and not all actions can be done directly**
- **Some tasks are better achieved through delegating**

  – e.g. spell checking
- **Can become screen space 'gobblers'**

- **Moving a mouse around the screen can be slower than pressing function keys to do same actions**

Which model is best?

- **Direct manipulation is good for 'doing' types of tasks,**

  e.g. designing, drawing, flying, driving, sizing windows

- **Issuing instructions is good for repetitive tasks, e.g. spell-checking, file management**
- **Having a conversation is good for children, computer- phobic, disabled users and specialised applications (e.g. phone services)**

Interaction and interface

- **Interaction type:**

  – what the user is doing when interacting with a system,

  e.g. instructing, talking, browsing or other

- **Interface type:**

  – the kind of interface used to support the mode,

  e.g. speech, menu-based, gesture

The 5 dimensions of interaction design:

1. **Words**
2. **Visual representations**
3. **Physical objects or space**
4. **Time**
5. **Behavior**